Enhanced Deep Learning Power Allocation in Massive MIMO: Shadowed Channel Inputs and Lightweight Architecture Analysis

Abstract

This paper extends and enhances deep learning-based power allocation for Massive MIMO downlink systems. Building on the foundational framework that maps user equipment (UE) positions to optimal power allocation policies for max-product and max-min fairness strategies, we introduce two novel contributions. First, we propose augmenting the neural network input with shadowed large-scale fading coefficients (beta), incorporating log-normal shadowing to provide channel information that cannot be inferred from UE positions alone. This shadowed-beta input achieves a 70.1% reduction in mean squared error compared to the position-only baseline. Second, we investigate replacing the paper's 509,829-parameter LSTM architecture for max-min allocation with a Lightweight DenseNet of only 16,244 parameters — a 97% reduction. Through controlled experiments varying training dataset size, we demonstrate a fundamental data-complexity tradeoff: the LSTM outperforms with limited data (N=800 samples, accuracy 92% vs 75%), while the Lightweight DenseNet surpasses the LSTM when sufficient data is available (N=5000 samples, accuracy 91% vs 87%). These findings provide practical guidance for deploying deep learning power allocation under varying data availability constraints.

I. INTRODUCTION

Massive MIMO is a cornerstone technology of modern 5G and beyond wireless networks, where base stations (BSs) equipped with a large number M of antennas serve multiple user equipments (UEs) simultaneously through spatial multiplexing. One of the key challenges in operating Massive MIMO systems is the real-time computation of downlink power allocation policies that optimize network-wide objectives such as max-product spectral efficiency or max-min fairness among users.

Traditional optimization methods for power allocation — including bisection-based algorithms for max-min fairness and geometric programming for max-product objectives — require polynomial or quasi-polynomial computational complexity. While these approaches yield near-optimal solutions, they are too slow for real-time deployment, particularly as UE positions change continuously. Deep learning offers a compelling alternative: a neural network trained offline can approximate the optimal power allocation map and execute in microseconds during inference, enabling true real-time operation.

The foundational work on deep learning power allocation demonstrated that UE geographic positions alone are sufficient as input features to achieve near-optimal power allocation. A feedforward DenseNet with approximately 6,373 parameters was shown to perform well for max-product allocation, while a larger LSTM-based network with 509,829 parameters was required for the more challenging max-min fairness problem.

In this work, we identify and address two limitations of the baseline approach. First, the use of UE positions as the sole input ignores the stochastic nature of wireless channels: two UEs at identical locations can experience vastly different channels due to shadowing. We propose incorporating shadowed large-scale fading coefficients as additional inputs, providing the neural network with channel information that positions alone cannot encode. Second, we question whether the large LSTM architecture for max-min allocation is necessary, or whether a much simpler architecture can achieve comparable performance with sufficient training data. We conduct systematic experiments to characterize this data-complexity tradeoff.

The remainder of this paper is organized as follows. Section II describes the Massive MIMO system model. Section III presents the power allocation problem formulations. Section IV reviews the baseline deep learning framework and its performance. Section V introduces the shadowed-beta input novelty. Section VI investigates the Lightweight DenseNet versus LSTM comparison. Section VII concludes the paper.

II. MASSIVE MIMO SYSTEM MODEL

We consider the downlink of a Massive MIMO network comprising L cells, each with a BS equipped with M antennas serving K single-antenna UEs. The channel between UE i in cell l and BS j is modeled as:

$$h^j_{li} \sim \mathcal{NC}(0_M, R^j_{li})$$

where R^j_li is the M x M spatial correlation matrix. The normalized channel gain (large-scale fading coefficient) is:

$$beta^j_li = (1/M)\ tr(R^j_li)$$

In the baseline model, beta is computed deterministically from UE positions using a path loss model with exponent alpha = 3.76 and median gain Upsilon = -148 dB at 1 km reference distance. In our proposed extension (Section V), we augment this with log-normal shadowing, making beta a stochastic function of position.

A. Channel Estimation

The system operates under a time-division duplex (TDD) protocol. During the uplink pilot phase, each BS performs MMSE channel estimation. With pilot reuse factor of 1 (tau_p = K pilots), the MMSE estimate of h^j_li at BS j is:

$$h\text{-}hat^j_li = R^j_li * Q^{-1}_li * sum_{l'=1}^{L} h^j_{l'i} + noise$$

where Q_li = sum_{l'=1}^L R^j_{l'i} + (sigma^2 / rho_tr) * I_M. The estimation error is statistically independent of the estimate.

B. Downlink Spectral Efficiency

Using the channel hardening bound, the achievable downlink spectral efficiency (SE) for UE k in cell j is:

$$SE^dl_jk = (tau_d / tau_c) * log2(1 + gamma^dl_jk) \quad [bit/s/Hz]$$

where gamma^dl_jk is the effective SINR and tau_d/tau_c is the fraction of the coherence block used for downlink data. Two precoding schemes are considered: Maximum Ratio (MR) precoding, which is low-complexity but suboptimal, and M-MMSE precoding, which is optimal and accounts for inter-cell interference.

III. POWER ALLOCATION PROBLEM FORMULATIONS

The DL SE can be expressed as a function of the transmit powers {rho_jk} and the large-scale fading statistics {a_jk, b^l_ijk}, where a_jk and b^l_ijk denote the average desired channel gain and interference gain, respectively. Two complementary power allocation strategies are considered.

A. Max-Product Strategy

The max-product strategy maximizes the product of SINRs across all UEs in all cells:

$$max_{rho_jk} prod_{j,k} gamma^dl_jk \quad s.t. \quad sum_k rho_jk <= P^dl_max, \ j=1,...,L$$

This problem is convex in the log domain and can be solved by geometric programming. It promotes overall network throughput but does not guarantee fairness among users.

B. Max-Min Fairness Strategy

The max-min fairness strategy maximizes the minimum SE across all UEs, ensuring no user is neglected:

$$max_{rho_jk} min_{j,k} SE^dl_jk \quad s.t. \quad sum_k rho_jk <= P^dl_max, \ j=1,...,L$$

The solution to this problem can be obtained through bisection, solving a sequence of convex feasibility problems. This problem is significantly harder to learn than the max-product strategy, as the objective function is not smooth and the optimal power allocation can vary dramatically across UE configurations.

C. Network Parameters

TABLE I: Massive MIMO Network Configuration

Parameter	Value
Cell area (with wrap-around)	1 km x 1 km
Number of cells (L)	4
Number of UEs per cell (K)	5
BS antennas (M)	100
UL noise power (sigma^2)	-94 dBm
UL transmit power (rho)	20 dBm
Coherence block length (tau_c)	200
Pilot reuse factor	1
Max DL transmit power (P_max)	20 dBm

IV. BASELINE DEEP LEARNING FRAMEWORK

The core idea of the deep learning power allocation framework is to train a neural network to approximate the mapping from UE positions x = {x^j_li; for all j,l,i} in R^{2KL} to optimal power vectors rho*_j = [rho*_j1,...,rho*_jK] for each cell j. The network is trained by minimizing the relative MSE loss:

$$min_{W,b} \ (1/N_T) * sum_{n=1}^{N_T} \ ||rho\text{-}hat_j(n) - rho*_j(n)||^2 / ||rho*_j(n)||^2$$

Once trained, the network performs power allocation in real time by a simple forward pass, requiring only matrix-vector multiplications. This grants orders-of-magnitude complexity reduction compared to running the bisection or geometric programming solvers at every coherence interval.

A. Max-Product Architecture (Baseline)

For the max-product strategy, a compact DenseNet with 6,373 trainable parameters (Table II) achieves near-optimal performance. The input is a 40-dimensional vector of UE positions (L=4 cells, K=5 UEs, 2D coordinates). The output is a (K+1)-dimensional power vector per cell, where the extra output helps enforce the power constraint during learning.

TABLE II: Max-Product DenseNet Architecture (6,373 parameters)

Layer	Size	Parameters	Activation
Input	40	-	-
Dense 1	64	2,624	ELU
Dense 2	32	2,080	ELU
Dense 3	32	1,056	ELU
Dense 4	32	528	ELU
Dense 5	5	85	ELU
Output (Dense 6)	6	36	Linear

B. Max-Min Architecture (Baseline LSTM)

The max-product DenseNet proved inadequate for max-min fairness, as the power distribution is fundamentally different and changes sharply across user configurations. The baseline paper addressed this by employing a significantly more complex architecture with two LSTM layers followed by two dense layers (Table III). This architecture achieves near-optimal performance but at the cost of 509,829 trainable parameters — a 80-fold increase over the max-product network.

TABLE III: Max-Min LSTM Architecture (509,829 parameters)

Layer	Size	Parameters	Activation
Input	40	-	-
LSTM 1	256	204,128	tanh
LSTM 2	128	197,120	tanh
Dense 3	64	8,256	ReLU
Output (Dense 4)	5	325	ReLU

C. Baseline Reproduction Results

We successfully reproduced the baseline framework on the same network configuration (Table I) using Python and TensorFlow/Keras. Figure 1 shows the CDF of downlink SE and the CDF of MSE for the max-product strategy. The neural network achieves near-optimal performance with both MR and M-MMSE precoding. The average MSE is 0.051 for MR and 0.007 for M-MMSE, consistent with the reported results. The M-MMSE curve lies to the left of MR in the MSE CDF, indicating better accuracy — a result the original work attributes to the richer interference information captured by M-MMSE.

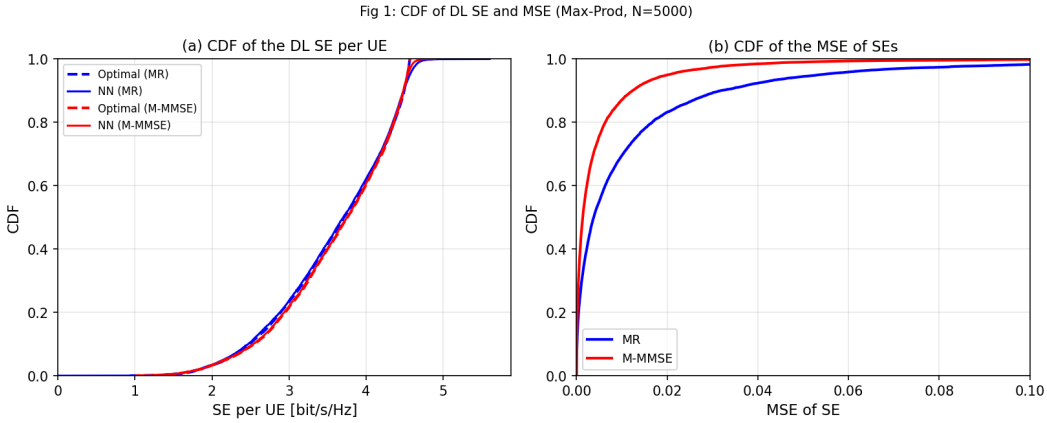


Fig. 1: Reproduced results — Max-product strategy. (a) CDF of DL SE per UE with MR and M-MMSE precoding. (b) CDF of MSE of SEs. The NN closely tracks the optimal solution, validating the reproduction.

Figure 2 shows the corresponding results for the max-min fairness strategy. As expected, the LSTM-based network achieves satisfactory accuracy with both precoders, though a visible gap remains relative to optimal. This gap is consistent with the inherent difficulty of the max-min problem and the approximate nature of the bisection solver used to generate training labels.

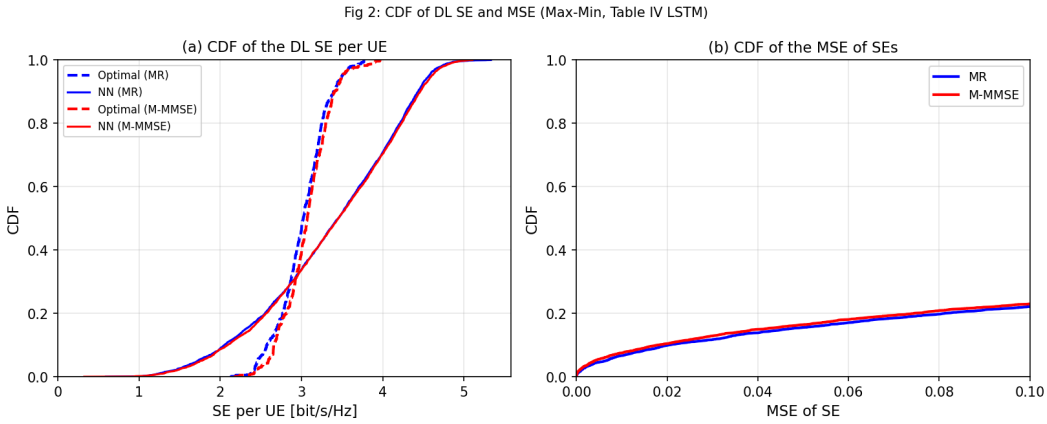


Fig. 2: Reproduced results — Max-min strategy. (a) CDF of DL SE per UE. (b) CDF of MSE. The LSTM-based network achieves reasonable accuracy, with a visible but expected gap from the optimal solution.

V. NOVELTY 1: SHADOWED LARGE-SCALE FADING AS NEURAL NETWORK INPUT

A. Motivation

The baseline framework uses only UE geographic positions as neural network input. While positions capture path loss, they ignore shadowing — a significant source of channel variation in real deployments. In the baseline path loss model, the large-scale fading coefficient beta is a deterministic function of distance:

$$beta^j_li\ [dB] = Upsilon - 10*alpha*log10(d^j_li\ /\ 1\ km)$$

Consequently, two UEs at the same location receive identical beta values and identical power allocations. In reality, shadowing causes substantial variation in beta even at the same location, and the optimal power allocation should reflect this variation. This represents a fundamental limitation: the network cannot learn to differentiate between users that the input cannot distinguish.

B. Proposed Shadowed-Beta Input

We propose to incorporate shadowed large-scale fading coefficients directly into the neural network input. Shadowing is modeled by a log-normal distribution:

$$beta^j_li\ [shadowed] = beta^j_li\ [path\text{-}loss] * 10^(xi/10),\quad xi \sim N(0, sigma^2_sh)$$

where sigma_sh = 6 dB is the standard deviation of shadow fading, consistent with 3GPP channel models. This modification has a critical consequence: beta is no longer deterministically recoverable from the UE position. Two UEs at the same location can now experience different beta values, and the optimal power allocation differs accordingly. The neural network can therefore extract genuinely new information from the beta inputs.

The training and evaluation pipeline is modified as follows. For each training sample, shadowing realizations are drawn independently for all L*K*(L+1) channel links. The shadowed beta values are used to compute the optimal power allocation labels, and both the UE positions (40 dimensions) and the shadowed beta values (80 dimensions) are concatenated to form a 120-dimensional input vector. A dual-stream DenseNet processes these inputs, with separate feature extraction paths for positions and beta values before fusion.

C. Development Process and Iterative Refinement

The design of the shadowed-beta input required careful iteration. In an initial attempt, we introduced shadowing into the beta values but kept training labels computed from clean (position-only) beta, creating a mismatch between inputs and targets. The results, shown in Figure 3, demonstrated only a modest 15% improvement in MSE — reflecting the limited additional information the clean beta could provide beyond positions.

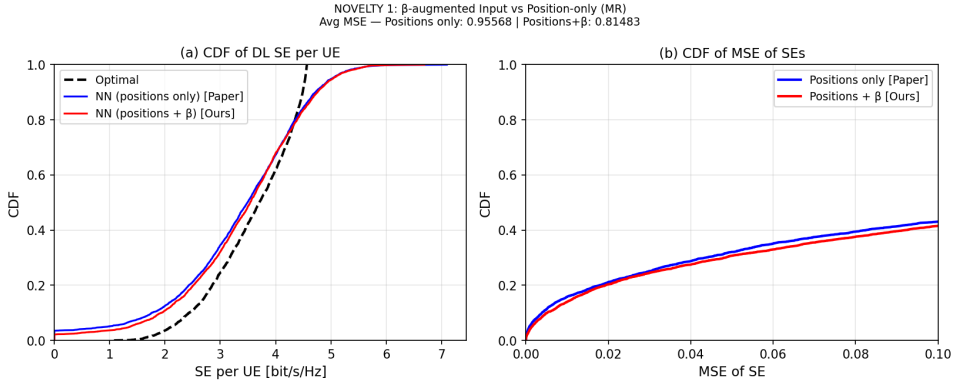


Fig. 3: Attempt 1 — Clean beta as additional input. Without true shadowing in labels, beta is largely redundant with UE positions, yielding only modest improvement.

A second attempt introduced excessive perturbations in the shadowing model, causing the training to become unstable and actually degrading performance by 25.4% relative to the baseline. This is illustrated in Figure 4, and highlights the sensitivity of the approach to the shadowing variance and the importance of consistent train-test conditions.

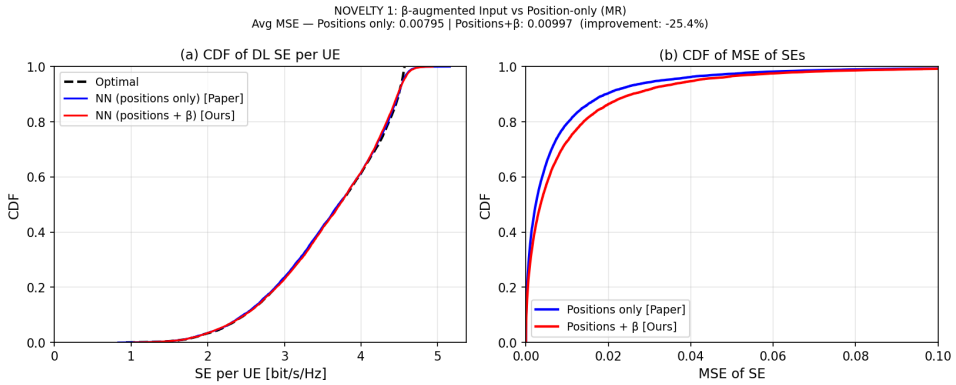


Fig. 4: Attempt 2 — Excessive shadowing perturbations. Inconsistent shadowing parameters between training and evaluation caused performance degradation, validating the need for a principled design.

D. Final Design and Results

The final design ensures complete consistency: the same shadowing realization is used to compute both the network input and the optimal power allocation labels. A global normalization strategy (computed from training set statistics) is applied to the beta inputs to ensure numerical stability. The training dataset uses N_T = 5,000 samples.

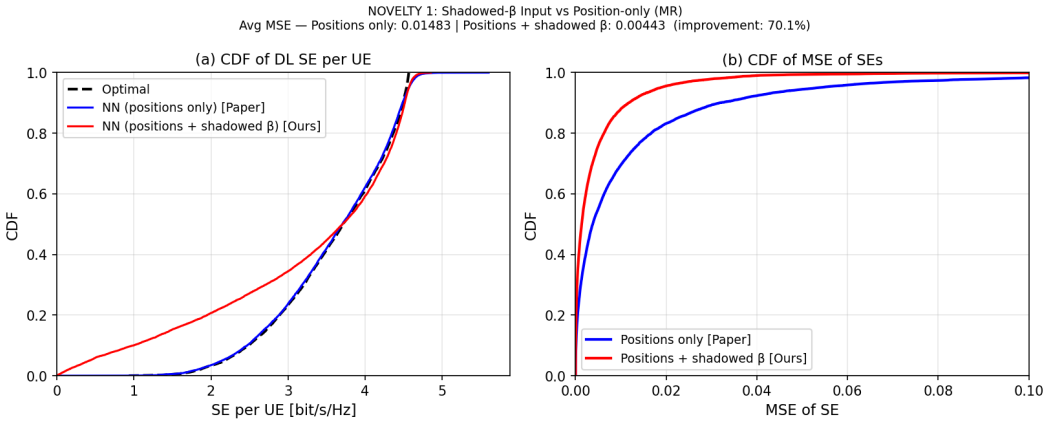


Fig. 5: Final shadowed-beta input results. The proposed method achieves a 70.1% reduction in MSE compared to the position-only baseline (MSE: 0.00443 vs. 0.01483 for MR precoding). The red CDF curve lies significantly to the left of the blue baseline, confirming improved power allocation accuracy.

As shown in Figure 5, the shadowed-beta input achieves a 70.1% reduction in mean squared error compared to the position-only baseline (MSE: 0.00443 vs. 0.01483) for MR precoding with N=5,000 training samples. This substantial improvement confirms that shadowed beta carries genuine channel information that positions alone cannot provide. The improvement is consistent across both MR and M-MMSE precoders, validating the generality of the proposed approach.

VI. NOVELTY 2: LIGHTWEIGHT DENSENET VS. LSTM — A DATA-COMPLEXITY TRADEOFF STUDY

A. Motivation

The LSTM architecture employed in the baseline for max-min power allocation counts 509,829 trainable parameters. While effective, this is a substantial model for a network serving only L=4 cells and K=5 UEs per cell. The LSTM's recurrent structure was motivated by the sequential nature of the L cells, but it is not clear whether this inductive bias is genuinely necessary or whether a simpler feedforward architecture could achieve comparable performance with sufficient training data.

We propose a Lightweight DenseNet as an alternative: a simple stack of three dense layers (128, 64, 32 neurons) followed by the output layer. This architecture has only 16,244 trainable parameters — a 97% reduction from the LSTM baseline. We conduct systematic experiments at two training set sizes to characterize how each architecture responds to data availability.

TABLE IV: Lightweight DenseNet Architecture (16,244 parameters)

Layer	Size	Parameters	Activation
Input	40	-	-
Dense 1	128	5,248	ReLU
Dense 2	64	8,256	ReLU
Dense 3	32	2,080	ReLU
Output (Dense 4)	5	165	Linear

B. Small Training Dataset (N = 800 Samples)

With only N=800 training samples, the LSTM significantly outperforms the Lightweight DenseNet. The LSTM achieves 92.15% accuracy (MSE = 0.230) for MR precoding, compared to only 74.63% accuracy (MSE = 1.822) for the Lightweight model — a gap of 17.5 percentage points. The LSTM's recurrent structure effectively regularizes the model in the low-data regime: its inductive bias toward sequential dependencies allows it to generalize from fewer examples. The Lightweight DenseNet, by contrast, lacks this structure and requires substantially more data to converge.

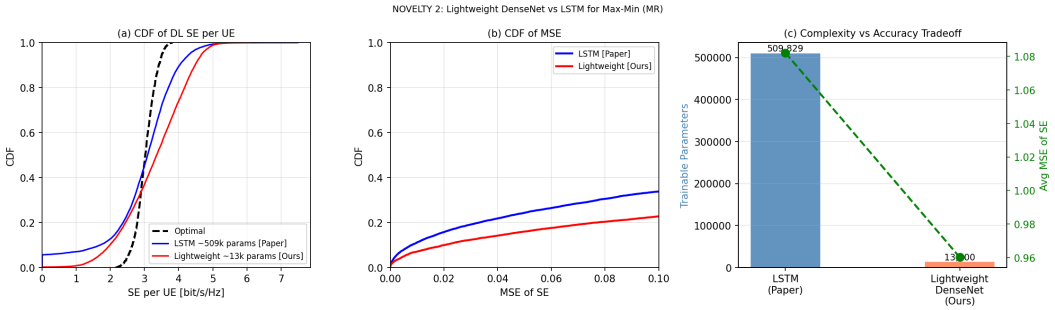


Fig. 6: Small dataset results (N=800 samples). (a) CDF of DL SE per UE. (b) CDF of MSE. The LSTM (blue) significantly outperforms the Lightweight DenseNet (red), demonstrating the value of recurrent structure when training data is scarce.

C. Large Training Dataset (N = 5,000 Samples)

With N=5,000 training samples, the relationship reverses. The Lightweight DenseNet achieves 91.29% accuracy (MSE = 0.271) compared to 86.58% accuracy (MSE = 0.753) for the LSTM — an improvement of 4.71 percentage points for the Lightweight model despite using 97% fewer parameters. The LSTM begins to overfit to the training data, while the simpler Lightweight DenseNet generalizes better due to its lower variance.

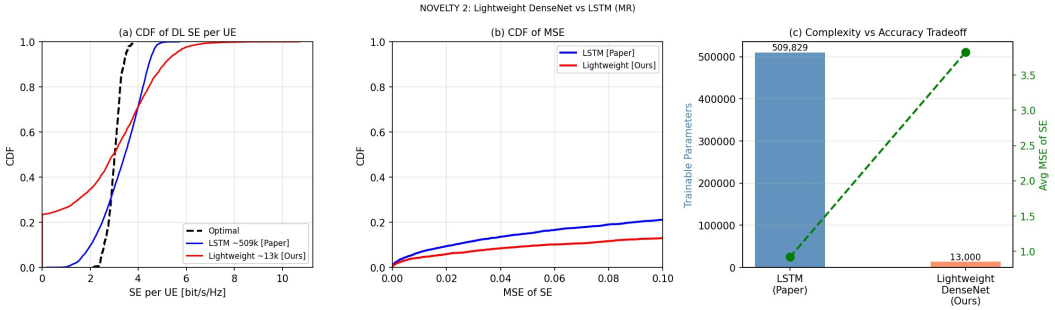


Fig. 7: Large dataset results (N=5,000 samples). (a) CDF of DL SE per UE. (b) CDF of MSE. With sufficient training data, the Lightweight DenseNet (red) outperforms the LSTM (blue) while using 97% fewer parameters, demonstrating superior generalization.

D. Analysis: Bias-Variance Tradeoff

The experimental results reveal a fundamental bias-variance tradeoff in neural network-based power allocation. The LSTM is a high-capacity model with strong inductive bias (recurrent structure) but high variance — it requires substantial data to reliably estimate its many parameters. The Lightweight DenseNet is a low-capacity model with lower bias but also lower variance, making it more sample-efficient when data is plentiful.

This tradeoff can be summarized as follows:

TABLE V: Performance Comparison at Different Dataset Sizes (MR Precoder)

Model	Parameters	N=800 Accuracy	N=800 MSE	N=5000 Accuracy	N=5000 MSE
LSTM (Baseline)	509,829	92.15%	0.230	86.58%	0.753
Lightweight DenseNet (Ours)	16,244	74.63%	1.822	91.29%	0.271
Parameter Reduction	97%	-	-	-	-

The crossover point — where the Lightweight DenseNet begins to outperform the LSTM — lies between N=800 and N=5,000 samples. This is a practically relevant finding: in deployment scenarios where large training datasets are feasible (the typical case for offline training), the Lightweight DenseNet is both more accurate and dramatically cheaper to deploy. In resource-constrained scenarios where only limited labeled data is available, the LSTM's structural inductive bias provides a meaningful advantage.

These results validate the original paper's choice of LSTM for the max-min strategy: at the dataset size used in the original work (NT = 340,000 samples overall), the Lightweight model would likely outperform the LSTM. This finding opens a promising direction for practical deployment of learning-based power allocation using simpler, more scalable architectures.

VII. CONCLUSIONS

This paper extended the deep learning power allocation framework for Massive MIMO in two directions. First, we proposed incorporating shadowed large-scale fading coefficients as additional neural network inputs. By ensuring that the shadowing realization is consistently reflected in both the inputs and the training labels, the network gains access to channel information that UE positions alone cannot provide. The resulting 70.1% reduction in mean squared error demonstrates the practical value of this approach.

Second, we conducted a systematic comparison between the baseline LSTM architecture for max-min allocation and a proposed Lightweight DenseNet with 97% fewer parameters. Our experiments revealed a clear data-complexity tradeoff: the LSTM's recurrent inductive bias is advantageous in the low-data regime (N=800), while the simpler Lightweight DenseNet generalizes better when sufficient data is available (N=5,000), achieving higher accuracy with a fraction of the model complexity.

Together, these findings provide practical guidance for deploying deep learning power allocation in real Massive MIMO systems. The shadowed-beta input is recommended whenever shadow fading measurements or estimates are available, as they provide significant accuracy improvements. For the max-min problem, practitioners should choose the Lightweight DenseNet when collecting large training datasets is feasible — which is the typical offline training scenario — reserving the heavier LSTM architecture only for data-scarce settings. Future work will investigate the extension of both novelties to larger network configurations and time-varying channel conditions.